

Full-Season SITS Inference on a Laptop in Under a Minute

BUCKET 7 - ON-DEVICE / EDGE-EFFICIENT INFERENCE

For agricultural extension officers who carry an Apple Silicon laptop rather than a server GPU, quantisation-aware distillation of a strong satellite-image-time-series transformer into a small student lets a full growing-season crop map over a co-op-sized area run locally in under a minute, exposing along the way that PyTorch's MPS backend has real, uncharted gaps for exactly this workload.

STRONG

This is the stronger of the two 'on-device/edge-efficient inference' candidates reviewed: its H1 novelty rests on a concrete, independently verifiable technical fact (PyTorch's INT8 quantization backend has no native MPS support) rather than a converging, crowded research area, its H6 pitch is highly demoable, and its main open items (naming a specific H4 user segment, resolving the MPS-vs-CoreML runtime ambiguity, and pre-registering accuracy-loss/latency targets) are straightforward fixes rather than fundamental weaknesses.

Confidence: Medium-High. The core novelty claim (no existing Apple-Silicon/MPS benchmarking of TSViT/SITSMamba/VistaFormer) and the key risk finding (MPS's missing native INT8 path) are both grounded in specific, checkable sources found this session (arXiv/OpenAlex searches across multiple query formulations, and community/vendor sources on PyTorch's MPS quantization gap). Confidence would rise with: a full-text read of AgriJetsonBench and the two Apple Silicon profiling papers to rule out any closer overlap; and an actual week-1 hands-on smoke test of the three named teacher models on MPS, which is exactly the first kill-criterion and has not yet been performed.

Hard constraint check

Constraint	Verdict	Reasoning
H1 • Novelty	CAVEAT	this session's searches (arXiv, OpenAlex, general web) found no existing paper benchmarking TSViT/SITSMamba/VistaFormer or any comparable full-season SITS model specifically on Apple Silicon/MPS; the closest adjacent work (AgriJetsonBench, Aug 2026) targets Jetson/CUDA single-frame agricultural vision models, not Apple Silicon SITS, making the gap plausible though not exhaustively confirmed by full-text review.
H2 • Not generic	PASS	not a banned generic pattern; a real systems+ML co-design contribution with a working demo, though the writeup must foreground the co-design (not just 'we ran a benchmark') to stay clearly outside a routine efficiency-ablation genre.
H3 • Imagery is primary input	PASS	Sentinel-1/2 SITS imagery is the entire input to both teacher and student models throughout.
H4 • Geography & user	CAVEAT	the candidate as described names only a generic 'extension officer'; H4 requires naming the actual user segment, so a specific department/co-op (e.g. Sarawak DOA, MADA Kedah, or an EU CAP-monitoring inspector) must be named explicitly in the final proposal.

H5 · Compute & budget**CAVEAT**

dataset and model sizes fit easily within 24GB and a free budget, but this session verified a real ambiguity: PyTorch's native INT8 quantization path is not available on MPS, so genuine quantization work likely requires CoreML/ANE export rather than staying in 'PyTorch MPS' as H5 literally states; this must be resolved and stated explicitly rather than discovered mid-project.

H6 · Pitch quality**PASS**

a strong, concrete, demoable pitch (live under-a-minute full-season inference on a laptop, unplugged) built around a genuinely verified hardware limitation that a CS judging panel can appreciate as non-trivial and not previously documented.

The problem

State-of-the-art SITS (satellite image time series) crop-mapping transformers — TSViT, SITSMamba, VistaFormer — are developed and benchmarked on server-class NVIDIA GPUs with CUDA. Nobody publishes what these models cost in latency, memory, or accuracy when run on the Apple Silicon hardware (or other low-power, no-CUDA hardware) that a field agronomist or extension officer would actually carry, and PyTorch's MPS backend is known to have operator-coverage and quantization-support gaps relative to CUDA that could silently break or slow down standard efficiency techniques. Current practice implicitly assumes a CUDA GPU is available at inference time, which is false for a laptop-only field/co-op deployment.

Proposed technical approach

Take one of the named SITS transformers (most likely VistaFormer or TSViT, given their manageable size) as a fixed teacher, pretrained on Sentinel-1/2 time series for parcel-level crop classification (PASTIS-R). Perform quantisation-aware distillation: train a smaller student architecture (fewer layers/heads, reduced hidden dim, or a lightweight CNN-temporal-encoder hybrid) using a combined loss of standard cross-entropy against ground truth plus a distillation term (KL divergence / feature-matching) against the teacher's soft outputs, while simulating quantisation noise during training (fake-quantisation / QAT-style weight and activation clamping) so the student is robust to the precision it will actually run at during deployment. Deployment path is chosen based on a week-1 smoke test: either (a) native PyTorch MPS with FP16/INT4 weight-only quantisation via torchao (confirmed to have some MPS support), or (b) export to CoreML via coremltools with CoreMLQuantizer for genuine INT8/Apple Neural Engine execution, since PyTorch's legacy INT8 quantized-tensor backend was never ported to MPS. The core empirical contribution is a full latency/memory/energy/accuracy characterization: measure wall-clock time and peak memory for a full growing-season inference pass over a fixed, co-op-sized area (e.g. one PASTIS-R patch or an equivalent-area Sentinel-1/2 stack) on the M4 Pro, at each precision/architecture point, and report accuracy-vs-latency and accuracy-vs-energy (measured via macOS powermetrics) Pareto fronts against the uncompressed teacher and against the three named baseline architectures run at their default (unoptimized) settings.

Novelty

WHAT THE DELTA ACTUALLY IS

The defensible delta is a genuine, previously undocumented deployment constraint — no CUDA, Apple unified memory, full-season (not single-frame) temporal context — applied to SITS crop-mapping transformers, which is materially different from the existing edge-inference literature that optimises single-frame detection/segmentation on Jetson-class CUDA devices (e.g. AgriJetsonBench, 2026, benchmarks crop/weed detection on Jetson AGX Orin/Orin Nano via TensorRT, not on Apple Silicon and not on multi-temporal SITS). This session's searches found no existing paper benchmarking TSViT/SITSMamba/

VistaFormer (or any comparable SITS model) on Apple Silicon/MPS, and confirmed a concrete, verifiable technical fact that strengthens the novelty claim: PyTorch's legacy INT8 quantized-tensor backend was never ported to MPS, meaning a standard CUDA-style QAT/PTQ INT8 workflow will not run as-is on this hardware and must route through CoreML/ANE conversion or MPS-specific workarounds (fake quantisation, torchao INT4 weight-only) — exactly the kind of 'MPS lacks kernels that CUDA implementations assume' obstacle the proposal names, now evidenced rather than assumed. This is **NOT** 'applied X to Y hardware' in the trivial sense: the quantisation-aware distillation recipe must be co-designed with the actual MPS/CoreML operator constraints, and the honest operator-coverage-and-throughput study is itself a first-class, previously-missing empirical contribution.

PRIOR ART THAT THREATENS IT

General-purpose (non-SITS) quantization-aware distillation frameworks for edge deployment already exist and would need to be positioned against, not ignored: EdgeRazor (arXiv:2605.04062, 2026, mixed-precision QAT-distillation for LLMs), and earlier CV/edge-AI deployment-strategy benchmarking studies by Singh et al. (arXiv:2403.17154 and arXiv:2411.00907, 2024) that empirically compare quantization, pruning, distillation, and partitioning operators across mobile/edge/cloud tiers on standard CV/NLP models (not SITS, not Apple Silicon specifically). AgriJetsonBench (arXiv:2608.00927, Aug 2026) is the closest agricultural-domain precedent: it rigorously benchmarks crop/weed vision models on Jetson edge GPUs with power-referenced timing, but targets CUDA-based Jetson hardware and single-frame detection/segmentation, not Apple Silicon or full-season temporal SITS — a real but adjacent, not identical, prior study, and a template this proposal should explicitly differentiate itself from in its writeup. The two Apple Silicon profiling papers found (Benazir & Lin 2025, arXiv:2508.08531; Feng et al. 2025, arXiv:2501.14925) are LLM/training-focused, not vision/SITS-focused, but their findings (quantization is not automatically faster on Apple Silicon; unified memory creates a different training performance profile than CUDA) are directly load-bearing risks for this proposal's core assumption and must be addressed rather than assumed away.

GENERIC-PATTERN CHECK

Confirmed not a chatbot/RAG assistant, dashboard, leaf-photo classifier, NDVI viewer, tabular yield model, plain benchmark re-run, or thin API-wrapper app. The core technical contribution — quantisation-aware distillation co-designed with a specific, verified hardware constraint (MPS's missing native INT8 path), evaluated as a genuine engineering artifact with latency/memory/energy Pareto fronts — is a real systems+ML contribution. Could the one-sentence summary describe 50 other FYPs? Less likely than the compression-focused sibling candidate: 'benchmark and compress a specific published model family for a specific, verified hardware gap, with a live working demo' is more distinctive than a generic 'make it fast on edge devices' framing, though the student must be careful the final writeup foregrounds the co-design (student architecture shaped by known MPS/CoreML constraints) rather than reading as a routine efficiency ablation.

Sensors and imagery

Sensor	Sentinel-1 (SAR, GRD, VV/VH) + Sentinel-2 (MSI, L2A) — same multi-sensor SITS stack used by TSViT/SITSMamba/VistaFormer and the PASTIS-R benchmark
Spatial resolution	10 (common resampled grid) m
Revisit	Sentinel-1: 6-12 days; Sentinel-2: ~5 days combined constellation
Bands used	Sentinel-1 VV/VH backscatter; Sentinel-2 10 optical bands as used in PASTIS-R
Licence	Copernicus open data licence (free, attribution only)

Yes. The entire teacher and student model input is the Sentinel-1/2 season time-series stack; the contribution is about how efficiently that imagery is processed on-device, not about adding any non-imagery input.

Datasets

PASTIS-R

Access route	Zenodo (via github.com/VSainteuf/pastis-benchmark), direct zip download, no auth
Auth required	No
Licence	CC (research use) per PASTIS-R repository
Approx. size	54 GB (full raster) or 27 GB (pixel-set format)
Accessible from MY	Yes, no geographic restriction found
How verified	WebSearch confirmed via GitHub repository description of file sizes and Zenodo hosting; this is the same dataset used by TSViT, SITSMamba, and VistaFormer's own papers, making it the natural shared benchmark for teacher/student/baseline comparison

MTLCC (MUNICH / LOMBARDIA CROP-SEGMENTATION BENCHMARK)

Access route	Referenced in the TSViT/VistaFormer comparative literature (Gatti et al. 2024, arXiv: 2412.01944) as a second SITS benchmark; exact hosting URL not independently fetched this session
Auth required	[uncertain]
Licence	[uncertain]
Approx. size	[uncertain] — expected small-to-moderate given it predates PASTIS-R in common usage
Accessible from MY	[uncertain]
How verified	Identified via arXiv abstract of the comparative study (2412.01944) referencing 'Munich and Lombardia datasets'; not independently verified this session, would need direct confirmation before relying on it

TOTAL DATA VOLUME

27-54 GB using PASTIS-R alone as the single primary benchmark for teacher training, student distillation, and all baseline comparisons — comfortably under the ~200GB H5 ceiling with no need for a second dataset.

LABELS AND GROUND TRUTH

PASTIS-R provides real parcel-level crop-type labels (18 classes) sourced from the French agricultural parcel register, giving high-quality, publicly verifiable ground truth for both teacher pretraining and student evaluation. No pseudo-labeling or weak supervision is required since the contribution is about efficient inference, not new-region label scarcity — this sidesteps the label-scarcity risk that the compression-focused sibling candidate (Bandwidth-Aware Satellite Inference) faces for a Malaysia case study.

Baselines to beat

Paper	Reported metric	Dataset
ViTs for SITS: Vision Transformers for Satellite Image Time Series (TSViT) arXiv / CVPR 2023 2023 · arXiv:2301.04944	State-of-the-art accuracy/mIoU on PASTIS and other SITS benchmarks, full-precision, server-GPU inference only	PASTIS and other public SITS datasets
SITSMamba for Crop Classification based on Satellite Image Time Series arXiv 2024 · arXiv: 2409.09673	Competitive crop classification accuracy while addressing the quadratic self-attention complexity of Transformer-based SITS models via a Mamba temporal encoder	SITS crop classification benchmarks (per paper)
VistaFormer: Scalable Vision Transformers for Satellite Image Time Series Segmentation arXiv 2024 · arXiv: 2409.08461	Improves PASTIS mIoU by 0.1% and MTLCC mIoU by up to 3.7% over prior SOTA while using only 8% (MHSA) or 11% (Neighbourhood Attention) of the FLOPs of comparable transformers — already the most efficiency-oriented of the three named teachers, and the most realistic starting point for further compression	PASTIS, MTLCC
A Comparative Study of Transformer and Convolutional Models for Crop Segmentation from Satellite Image Time Series arXiv 2024 · arXiv: 2412.01944	Head-to-head comparison of 3D U-Net, 3D FPN, 3D DeepLabv3, Swin UNETR, TSViT, and VistaFormer on Munich/Lombardia Sentinel-2 data; confirms TSViT/VistaFormer as the strongest available SITS teachers and VistaFormer as the efficiency-optimal choice	Munich, Lombardia (Sentinel-2)
Profiling Large Language Model Inference on Apple Silicon: A Quantization Perspective arXiv 2025 · arXiv: 2508.08531	Benchmarks M2 Ultra, M2 Max, and M4 Pro against NVIDIA GPUs across 14 quantization schemes; explicitly finds that lower-bit compression is not a universal speed guarantee on Apple Silicon due to dequantization overhead — a direct, load-bearing risk finding for this proposal's core assumption	N/A (systems profiling study, LLMs not SITS)

Evaluation protocol

Primary metrics: (1) macro-F1 / mIoU for crop classification on PASTIS-R's official 5-fold cross-validation test folds; (2) wall-clock latency (seconds) for a full growing-season inference pass over one fixed-size co-op-representative area (e.g. one PASTIS-R patch, ~1.28km x 1.28km) on the M4 Pro; (3) peak resident memory (GB); (4) energy draw (measured via macOS powermetrics) during that inference pass. Report accuracy-vs-latency and accuracy-vs-energy Pareto fronts comparing: the uncompressed named teachers (TSViT, SITSMamba, VistaFormer) run as-is on MPS, the distilled-only student (no quantisation), and the distilled+quantised student (final proposed system). Success criterion, to be pre-registered before building: full-season inference completes in under 60 seconds over the fixed area with no more than a stated acceptable macro-F1 drop (e.g. 3-5 points) relative to the best uncompressed teacher on the same fold.

Feasibility

COMPUTE

Peak memory	Estimated 4-8 GB for training a VistaFormer/TSViT-scale teacher on PASTIS-R patches (moderate batch size), and 2-5 GB for the smaller distilled student; both comfortably under 24 GB unified memory
Est. training hours (MPS)	Estimated 4-10 hours per teacher-training run and a further several hours per distillation/QAT configuration on the M4 Pro; per-run time likely 1.5-3x slower than an equivalent CUDA GPU based on the Apple Silicon training-profiling literature (Feng et al. 2025), so a full multi-configuration sweep (student sizes x precision levels) may exceed a single 12-hour session and require spreading across several sessions or offloading teacher pretraining to free Colab/Kaggle T4 quota
Fits 24 GB	Yes for all individual training/inference runs; the binding constraint is wall-clock time across the full experiment matrix, not memory
MPS-compatible ops	CONFIRMED (verified via WebSearch this session, not assumed): PyTorch's legacy torch.quantization INT8 quantized-tensor backend was never ported to the MPS backend. Practical routes are (a) 'fake quantization' — store weights at low bit-width, dequantize to FP16 at matmul time — which does run with GPU acceleration on MPS; (b) torchao, which is adding native INT4 weight-only quantization support for MPS; or (c) exporting via coremltools to CoreML and using CoreMLQuantizer for genuine 8-bit static quantization targeting the Apple Neural Engine, which is a different runtime/toolchain than 'PyTorch on MPS' as literally stated in H5 and should be clarified in the proposal.
Colab/Kaggle fallback	Likely yes for teacher pretraining and the full precision/architecture sweep, even though the final on-device demo and MPS-specific benchmarking must happen on the M4 Pro itself (that IS the point of the study)
Notes	This is the one place in either candidate where a concrete, previously-undocumented (to this session's searches) hardware limitation was found and verified rather than merely assumed, which is a genuine strength for the 'honest MPS operator-coverage study' framing of the pitch.
Cash cost	Approximately RM 0. PASTIS-R is free (Zenodo), no commercial imagery needed, all software (PyTorch, coremltools, torchao) is free and open-source. Optional Colab Pro (~RM45/month) could shorten the teacher-pretraining/sweep wall-clock time but is not mandatory given free-tier Colab/Kaggle T4 quota.
Build : research	Roughly 75:25 engineering-to-research. Engineering: implementing the quantisation-aware distillation training loop, the MPS-vs-CoreML export/deployment toolchain, a latency/memory/energy measurement harness (powermetrics integration), and the live demo application. Research: characterizing the actual MPS/CoreML operator-coverage gaps for this model family, designing the student architecture around those constraints, and honestly reporting the accuracy-efficiency Pareto front including where quantisation does NOT help (per the Benazir & Lin 2025 finding). This matches the stated S2 preference for a build-heavy project with a working demo.
Publication venue fit	IGARSS (a systems/efficiency-oriented track fits its scope) or the ICLR ML4RS workshop are the most realistic venues given the compression + remote-sensing crossover and the emphasis on a reproducible benchmarking contribution; Remote Sensing (MDPI) is a plausible journal fallback given its history of publishing SITS efficiency studies.

DEMO ARTIFACT

Live laptop demo: a full growing-season Sentinel-1/2 stack over a fixed, stated co-op-sized area is fed to the distilled+quantised student running natively on the M4 Pro (unplugged from power to also show battery/energy draw), with an on-screen stopwatch counting down and a crop map rendering as it completes in under a minute; a side-by-side panel shows the same area processed by the unoptimized teacher model taking substantially longer, making the speed/memory/accuracy tradeoff immediately visible to a judging panel.

User and pain

USER SEGMENT

Agricultural extension officers and agronomists working for Malaysian state agriculture departments (e.g. Sarawak Department of Agriculture, or Kedah's Muda Agricultural Development Authority) who visit co-ops and smallholder farms carrying a personal or department-issued laptop, needing an in-field, offline crop-map/season-progress check without depending on a data connection or a returned trip to an office GPU workstation. (The candidate description as given names only a generic 'extension officer'; H4 requires naming the actual segment, so this should be sharpened in any proposal document, e.g. to a specific department or an EU CAP-monitoring inspector doing on-farm eco-scheme verification visits as an EU-segment alternative.)

QUANTIFIED PAIN

A quantified pain number specific to 'extension officers carrying laptops in the field' was not found this session and should be treated as a genuine gap: the pitch's implicit pain (no GPU/no connectivity during a farm visit, so cloud-server inference is unavailable) is plausible but currently unquantified. A usable proxy, not independently verified this session: Malaysia's Department of Agriculture reportedly conducts periodic physical crop-condition/subsidy-verification visits to paddy schemes (e.g. under MADA in Kedah) where officers currently rely on manual/visual assessment rather than per-visit data-driven crop maps — this framing needs a real citation (e.g. MADA or DOA annual report) before being presented to a panel as a hard number.

Key risks

1. Novelty risk: while no directly matching SITS-on-Apple-Silicon paper was found this session, general-purpose edge-AI quantization-distillation frameworks (EdgeRazor, torchao, coremltools quantization) already exist, so the contribution rests on the specific combination (SITS temporal models + Apple Silicon + full-season context) rather than a new mechanism — a full-text literature recheck in week 1 is warranted, not just an abstract-level search.
2. Accuracy risk: VistaFormer is already highly FLOP-efficient (8-11% of comparable transformers), so further distillation+quantisation may hit diminishing returns or unacceptable accuracy loss on an already-compact teacher; TSViT or SITSMamba (larger, less optimized) may be a better distillation starting point but were not benchmarked for on-device efficiency in their own papers.
3. Toolchain/framework risk: genuine INT8 quantisation gains on Apple hardware likely require routing through CoreML/Apple Neural Engine rather than staying purely in 'PyTorch MPS' as H5 literally specifies; this is a real ambiguity in the H5 constraint that should be resolved explicitly (is the target runtime PyTorch-MPS, or CoreML/ANE, or both compared honestly?) before committing to an architecture.
4. Measurement risk: laptop-based energy/latency measurement (accuracy-per-watt, accuracy-per-second) is sensitive to thermal throttling, background processes, and battery vs. plugged-in state; without a controlled measurement protocol (consistent thermal conditions, powermetrics sampling, repeated trials) the headline Pareto front could be noisy or non-reproducible.

5. Scope-definition risk: 'co-op-sized area' and 'under a minute' are currently undefined; without pre-registering concrete numbers (hectares, seconds) early, the success criterion can silently drift to whatever the final system happens to achieve.

Kill criteria

1. Week 1: run the three named teacher architectures (TSViT, SITSMamba, VistaFormer) as-is, unmodified, as a forward pass on MPS over a representative PASTIS-R patch; record whether each runs at all, and its baseline latency/memory. If none run without extensive custom kernel rewriting, rescope which teacher(s) to use.
2. Week 1: attempt an actual INT8 (via CoreMLQuantizer/CoreML export) or INT4 weight-only (via torchao on MPS) quantisation of the smallest teacher and measure the real latency/memory delta versus FP32/FP16. If quantisation yields no measurable speed benefit (consistent with the Benazir & Lin 2025 finding on Apple Silicon), pivot the efficiency story toward distillation-driven size/memory reduction rather than quantisation-driven speed.
3. Week 2: pre-register the exact 'co-op-sized area' (in hectares or km^2) and the 'under a minute' latency target as concrete numbers, and confirm a full season of Sentinel-1/2 tiles for that area can realistically be prepared/cached within the compute and time budget.
4. Week 2: measure the distilled+quantised student's macro-F1 against the best uncompressed teacher on a PASTIS-R held-out fold; if the drop exceeds a pre-registered acceptable threshold (e.g. 5 F1 points), treat it as a kill/pivot signal per the task's explicit 'verify the accuracy loss is acceptable' instruction.

Open / unverified items

- Whether MTLCC (Munich/Lombardia) dataset access route, size, and licence are actually as assumed — not independently fetched this session, referenced only via a comparative study's abstract
- A real, citable quantified-pain statistic for extension officers/agronomists doing offline in-field crop assessment in Malaysia or the EU — not found this session, currently a plausible but unverified pain claim
- Whether a full-text read of AgrijetsonBench or the two Apple Silicon profiling papers would reveal closer overlap with this proposal than the abstracts alone suggest
- The exact wall-clock training time for a SITS transformer teacher and its distilled student on M4 Pro MPS specifically — estimated by analogy to general Apple Silicon vs CUDA profiling literature, not directly measured
- Whether torchao's MPS INT4 weight-only quantization or CoreML's CoreMLQuantizer INT8 path will integrate cleanly with a custom SITS transformer architecture (as opposed to the standard vision/language architectures these tools are most commonly demonstrated on) — this is precisely the week-1 smoke test, not yet performed

Key references

ViTs for SITS: Vision Transformers for Satellite Image Time Series (TSViT)

arXiv / CVPR 2023 · 2023 · arXiv:2301.04944

One of the three named teacher models explicitly cited in the pitch; the full-precision, server-GPU SOTA baseline to be compressed and benchmarked on MPS.

SITSMamba for Crop Classification based on Satellite Image Time Series

arXiv · 2024 · arXiv:2409.09673

Second named teacher; a Mamba-based alternative to attention that may have a different (possibly more MPS-friendly, possibly less) operator profile worth characterizing.

VistaFormer: Scalable Vision Transformers for Satellite Image Time Series Segmentation

arXiv · 2024 · arXiv:2409.08461

Third named teacher and already the most FLOP-efficient of the three (8-11% of comparable transformer FLOPs), making it both the most realistic distillation starting point and the model most likely to show diminishing returns from further compression.

Profiling Large Language Model Inference on Apple Silicon: A Quantization Perspective

arXiv · 2025 · arXiv:2508.08531

Directly evidences the core risk that quantization is not automatically faster on Apple Silicon due to dequantization overhead — a finding the proposal must confront rather than assume away, and the closest existing systematic Apple Silicon hardware-profiling methodology to emulate.

Profiling Apple Silicon Performance for ML Training

arXiv · 2025 · arXiv:2501.14925

Explains the unified-memory-driven training performance gap between Apple Silicon and NVIDIA GPUs, directly informing the `est_training_hours_mps` risk and time-budgeting for the teacher-training/distillation sweep.

AgriJetsonBench: External-Power-Referenced TensorRT Benchmarking of Agricultural Vision Models on Jetson Edge Platforms

arXiv · 2026 · arXiv:2608.00927

Closest adjacent agricultural-domain edge-benchmarking precedent (power-referenced, deployment-realistic methodology) but targets Jetson/CUDA single-frame vision models, not Apple Silicon/full-season SITS — an essential paper to differentiate against explicitly in the writeup.

A Comparative Study of Transformer and Convolutional Models for Crop Segmentation from Satellite Image Time Series

arXiv · 2024 · arXiv:2412.01944

Independently confirms TSViT and VistaFormer as the strongest available SITS teachers and provides a second benchmark dataset (Munich/Lombardia) beyond PASTIS-R for potential cross-dataset generalisation checks.

Generated from the source proposal JSON. Amber [uncertain] flags are carried over verbatim from the source and mark claims that were not independently verified.